

A. BACKGROUND SIGNAL

Timestamp + location
(t, s)

External atmospheric product
or synchronized monitoring network

Background estimate
 $B(t, s)$

Independent of the target observation being predicted

B. VISUAL LOCAL-INCREMENT BRANCH

$T = 7$ image sequence

ResNet50
visual embeddings

Temporal encoder
GRU / conditioned GRU

Visual local estimate
 $\hat{\Delta}_{visual}$

C. STRUCTURED CONTEXT BRANCH

Available atmospheric, temporal,
traffic, road and site context

Tree-based
local model

Context local estimate
 $\hat{\Delta}_{context}$

D. VALIDATION-SELECTED FUSION & OUTPUT

Validation-selected fusion
 $F(\hat{\Delta}_{visual}, \hat{\Delta}_{context})$

Fused local increment
 $\hat{\Delta}_{local}$

Add background
 $B + \hat{\Delta}_{local}$

Final PM_{2.5} prediction
 $\hat{y}(t, s)$

